

ELVIS TANG

Embedded Systems | Firmware Engineering | C/C++

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Computer Engineering graduate with hands-on experience developing embedded C/C++ software for sensor-fusion and autonomous systems. Experienced with UART, SPI, I2C, Raspberry Pi, Arduino, real-time data acquisition, hardware/software integration, system testing, and technical documentation.

EXPERIENCE

Embedded Systems Project Lead - Lockheed Martin-Sponsored Capstone

Aug 2022 - May 2023

California State Polytechnic University, Pomona

Pomona, CA

- Developed and debugged embedded C/C++ software integrating LiDAR, stereo camera, IMU, ultrasonic sensors, and wheel encoders on an autonomous ground vehicle.
- Implemented UART, SPI, and I2C communication on Raspberry Pi and Arduino; applied Kalman filtering to real-time sensor data for obstacle avoidance.
- Tested sensor performance across varied surfaces, validating distance measurements within +/-10 cm and real-time object classification up to 50 cm.

Instructor

Nov 2023 - Sep 2024

Coding Minds Academy

Irvine, CA

- Taught Python, JavaScript, Git, and introductory hardware concepts to classes of 10+ students, connecting programming fundamentals with hands-on engineering.
- Guided software and hardware projects through design, implementation, debugging, and demonstration.
- Developed project-based curricula emphasizing programming logic, hardware concepts, and practical troubleshooting.

Stand Lead

Aug 2024 - Present

Aramark

Anaheim, CA

- Patch and troubleshoot point-of-sale systems to maintain reliable register operation during high-volume events.
- Coordinate concession operations serving 1,000+ guests per shift while resolving 20+ customer or operational issues.
- Manage inventory for 50+ items and follow cash-handling, food-safety, and workplace procedures in a fast-paced environment.

SELECTED PROJECTS

Sensor Fusion and Tracking for an Autonomous Ground Vehicle | C/C++, Raspberry Pi, Arduino, Kalman Filtering

- Integrated LiDAR, stereo camera, ultrasonic sensors, IMU, and wheel encoders into a multi-sensor perception and tracking architecture.
- Implemented UART, SPI, and I2C interfaces across Raspberry Pi, Arduino, and sensor components for real-time data acquisition.
- Fused real-time sensor data with Kalman filtering to support obstacle detection, localization, and autonomous navigation.
- Validated performance across varied surfaces with distance measurements within +/-10 cm and object classification up to 50 cm.

LucidGloves - VR Haptic Glove Technical Build Guide | ESP32, Haptics, 3D Printing, SteamVR

- Produced an illustrated implementation guide for an open-source VR haptic glove covering 3D-printed components and mechanical assembly.
- Documented electronics and wiring steps to support repeatable assembly and hardware/software integration.
- Explained ESP32 firmware setup, SteamVR calibration, and end-to-end troubleshooting for the completed glove.
- Analyzed sensing, actuation, and haptic-feedback concepts that translate hand motion into responsive virtual-reality interaction.

Processor Architecture Performance Comparison | SilverBench, Benchmarking, Computer Architecture

- Ran repeatable SilverBench workloads to compare processor behavior under a consistent test methodology.
- Collected performance signals and documented the test configuration to establish context for the comparison.
- Interpreted results through computational throughput, workload behavior, and architecture-level tradeoffs.
- Presented observations, limitations, and conclusions in a structured technical report.

Traffic Crash Data Research | R, Logistic Regression, Statistical Modeling

- Analyzed five years of Highway Safety Information System data covering driver, roadway, vehicle, and environmental factors.
- Recoded categorical variables and modeled hit-and-run associations using logistic regression in R.
- Applied backward variable selection using Akaike Information Criterion (AIC) to retain relevant predictors and limit overfitting.
- Reviewed residual and leverage diagnostics; reported severity, lighting, and intersection population associations.

EDUCATION & CERTIFICATIONS

B.S., Computer Engineering - California State Polytechnic University, Pomona

Certification: CompTIA Security+ (SY0-701)

TECHNICAL SKILLS

Firmware & Software: C/C++, Python, embedded debugging, real-time data acquisition, UART, SPI, I2C

Platforms & Hardware: Raspberry Pi, Arduino, ESP32, LiDAR, stereo camera, ultrasonic sensors, IMU, wheel encoders

Engineering Methods & Tools: sensor fusion, Kalman filtering, obstacle avoidance, hardware/software integration, Git, GitHub, MATLAB